Timing of Intrapartum Ampicillin and Prevention of Vertical Transmission of Group B Streptococcus

MARINA DE CUETO, MD, PhD, MARIA-JOSE SANCHEZ, MD, ANTONIO SAMPEDRO, PhD, JOSE-ANTONIO MIRANDA, MD, PhD, ALFONSO-JESUS HERRUZO, MD, PhD, AND MANUEL ROSA-FRAILE, MD, PhD

***Objective*: To evaluate the relationship between the time elapsed from the administration of ampicillin prophylaxis to delivery and its efficacy in interrupting intrapartum transmission of group B streptococcus.**

***Methods*: During the 12-month study period, all women who came to the Virgen de las Nieves Hospital (Granada, Spain) for delivery were screened for group B streptococcus vaginal carriage by a pigment-detection culture-based procedure. Colonized women were treated with ampicillin (2 g intravenously), and the interval between ampicillin administration and delivery was recorded. Newborns from colonized mothers also were screened to detect group B streptococcus colonization.**

***Results*: During the study period, 4525 women were admitted to the hospital for delivery and screened for group B streptococcus vaginal colonization. Group B streptococcus was detected in 543 women (12%), of whom 454 gave birth vaginally to 454 liveborn infants. Intrapartum ampicillin was given to 201 of these 454 women (44%), and 10% of the newborns from mothers who received intrapartum ampicillin prophylaxis were colonized by group B streptococcus. The relationship between timing of ampicillin administration and rate of neonatal group B streptococcal transmission was as follows: less than 1 hour before delivery, 46%; 1–2 hours, 29%; 2–4 hours, 2.9%; and more than 4 hours, 1.2%. Among the 253 mothers who received no intrapartum prophylaxis, colonization was found in 120 of their newborns (47%).**

***Conclusion*: When the time between the start of ampicillin prophylaxis and delivery is at least 2 hours, vertical transmission of group B streptococcus is minimized. (Obstet Gynecol 1998;91:112–4.**

From the Microbiology Service and the Department of Obstetrics and Gynaecology, Virgen de las Nieves Hospital, Granada, Spain.

Group B streptococcus, or *Streptococcus agalactiae*, is an important cause of neonatal sepsis.[1] Most cases of neonatal infection occur by vertical transmission from a vaginally colonized mother to her infant during labor. It has been established that intrapartum ampicillin or penicillin given to group B streptococcus-infected mothers decreases vertical transmission and therefore reduces the incidence of early-onset disease.[2–4]

Ideally, ampicillin or penicillin should be administered at least 4 hours before delivery to achieve optimal concentrations in the amniotic fluid and the placental circulation.[5] However, this may be impractical under some circumstances, such as short duration of labor, parturients presenting in advanced labor, or unknown group B streptococcus carrier status at hospital admission.[6,7] The purpose of this study was to analyze the relationship between the time elapsed from ampicillin administration to delivery and its efficacy in preventing intrapartum vertical transmission of group B streptococcus.

Materials and Methods

The clinical trial from which these data were derived was done at the Virgen de las Nieves Hospital, a 1200-bed tertiary-care hospital in Granada, Spain, and has been described in detail elsewhere.[8] During March 1994 to February 1995, a culture was obtained by vaginal swab to detect group B streptococcus vaginal colonization from all 4525 women who came to our hospital for delivery. These cultures were taken on admission, in the obstetrics emergency room. Vaginal swab cultures were obtained from the distal vagina, without speculum placement, and were inoculated directly in tubes of the specific group B streptococcus

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pigment–enhancing medium[9] known as Granada medium (Biomedics, Madrid, Spain). These 4525 women gave birth to 4553 liveborn infants, which includes 50 twin pairs. There were 22 stillborns.

As soon as a positive culture result was observed (orange-red colonies), it was reported to the attending obstetric physician. Once it was known that a woman in labor was colonized with group B streptococcus, she was treated with ampicillin (2 g intravenously [IV] plus 1 g IV every 4 hours until delivery),[5] regardless of risk factors. Ampicillin administration was initiated at the onset of labor when a positive screening result was known at this time. Ampicillin administration started later, during labor, when a positive culture result was not known until after the onset of labor. Thus, the time between ampicillin administration and delivery varied from between 20 minutes and longer than 4 hours.

All newborns from colonized mothers were screened to detect colonization or infection by group B streptococcus. An aerobic blood culture bottle (BactT/Alert System; Organon Tecknica, Durham, NC) was inoculated, and urine, gastric content, and three mucocutaneous areas (pharynx, umbilicus, and external auditory canal) were cultured in tubes of Granada medium and plates of nalidixic blood agar.

Results

During the study period, group B streptococcus was detected in 543 women (12%), of whom 454 gave birth vaginally to 454 live newborns (no twin pairs). Positive samples for group B streptococcus were detected usually in tubes of Granada medium (red colonies) within 12 hours. However, because continuous 24-hour surveillance was not possible, only 65% of the positive cultures were reported within this period; after 18 hours 95% of all positive cultures were reported.[8,9]

Intrapartum ampicillin was given to 201 (44%) of the 454 colonized women who delivered vaginally. Twenty of the 201 newborns (10%) from mothers who received intrapartum ampicillin prophylaxis were colonized by group B streptococcus. No neonates from these 201 mothers who received intrapartum ampicillin developed group B streptococcal disease, and no adverse reactions were observed.

In the remaining 253 (56%) group B streptococcus–colonized mothers, intrapartum ampicillin prophylaxis was not given, because the test result was not available before the end of labor, or because of protocol violations. Among the 253 newborns from colonized mothers without intrapartum ampicillin prophylaxis, colonization was found in 120 newborns (47%), and one developed early-onset group B streptococcal disease.

The relation between the time elapsed from ampicillin administration to delivery and the rate of vertical transmission proved by positive cultures from newborns is shown in Table 1.

Table 1. Timing of Intrapartum Ampicillin Administration and Rate of Vertical Transmission

Time between ampicillin administration and delivery (h)	No. group B streptococci carriers	No. colonized newborns (%)
<1 h	24	11 (46)
1–2 h	21	6 (28)
>2–4 h	70	2 (2.9)
>4 h	86	1 (1.2)
Control group (no ampicillin)	253	120 (47)

Discussion

To date, the only practical way to decrease vertical transmission of group B streptococcus from a colonized mother to her newborn is to give intrapartum antibiotic prophylaxis to colonized mothers.[1,2,5] For pregnant group B streptococcal carriers, the Centers for Disease Control and Prevention (CDC)[5] recommends the administration of penicillin or ampicillin at least 4 hours before delivery. However, the optimal use of prophylaxis depends on the accurate identification of group B streptococcus–positive mothers at the time of labor.[5]

The CDC[5] recommends that pregnant women be screened for group B streptococcal colonization with vaginal cultures made close to the time of delivery (during weeks 36–38), because the results of these cultures are then available at hospital admission for delivery. However, previous vaginal cultures are not always done,[6,7] and the results of the culture may not always be available on hospital admission.

Antigen detection tests are not yet sensitive enough to detect group B streptococcus carrier women accurately on admission to the hospital, and therefore, these rapid tests should not be used to predict vaginal carriage of group B streptococcus.[2]

Even if intrapartum chemoprophylaxis were used perfectly, it still would not have eliminated all neonatal group B streptococcal infections, because prophylaxis is less than 100% effective. In addition, failure of intrapartum prophylaxis to interrupt vertical transmission of group B streptococcus could result from either inadequate dosage or inappropriate period for the antibiotic to reach significant levels in amniotic fluid, or from maternal genital secretions.[10]

Recent clinical reports[11,12] show that one of the specific problems in establishing a protocol for universal screening and prophylaxis for group B streptococcus is failure to administer antibiotics promptly. These reports[11,12] document a significant reduction in vertical transmission rate after intrapartum treatment of colonized mothers. However, the relationship between timing of antibiotic administration and its effectiveness has not been reported.

In a previous study,[8] we reported that intrapartum treatment decreased considerably the transmission of group B streptococcus from colonized mothers to their infants, from 47% to 10%. Nevertheless, when the time elapsed between ampicillin administration and delivery was analyzed, we observed that ampicillin was effective in interrupting vertical transmission of group B streptococcus only when it was administered at least 2 hours before delivery.

The present observation suggests that ampicillin prophylaxis should be administered to parturient women at least 2 hours before delivery to minimize group B streptococcus colonization of the newborn.

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Address reprint requests to:
Manuel Rosa-Fraile, MD, PhD
Servicio de Microbiologica
Hospital Virgen de las Nieves
Avda. Coronel Muñoz, 2
18014 Granada
Spain
E-mail: delarosa@cica.es

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